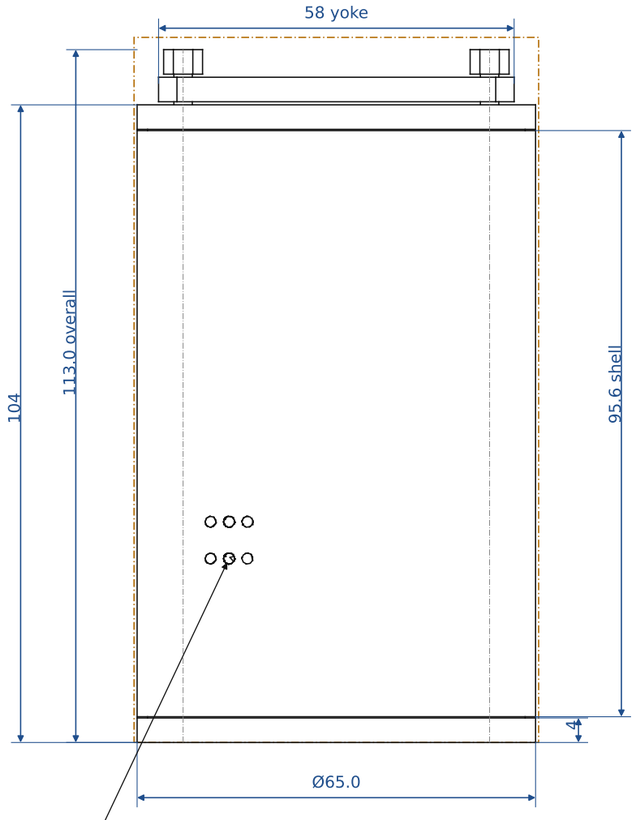
PROPOSED DESIGN — NOT YET FABRICATED. Values marked * are PRELIMINARY (module outlines are estimates to be measured on the actual parts when received).

Mechanical assumptions and unresolved dimensions

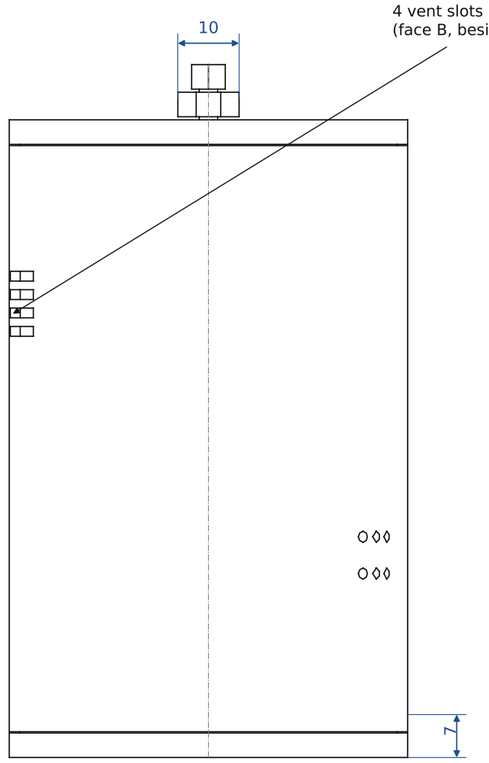
#	Assumption / unresolved item	Status
A1	Envelope Ø66 × 115 from ESA CanSat; SEVA FIRST publishes none	ASSUMED
A2	Rigid height excludes parachute, Y-bridle and swivel	ASSUMED
A3	PETG prints, 0.4 mm nozzle; spigot fit 0.1 mm radial	PROPOSED
A4	Module outlines from listings, not drawings	PRELIM.*
A5	JST-XH mated height at carrier edge ≈2.4 mm to bore	TO VERIFY
A6	Shell held by spigots (0.4 axial float); anti-rotation	UNRESOLVED
A7	Final mass / ballast need measured masses	UNRESOLVED

Load path (recovery): yoke → 2 × M3 A2 rods → bottom cap nut pockets; the spine is clamped between the caps and carries the modules; the shell is a non-structural sleeve.

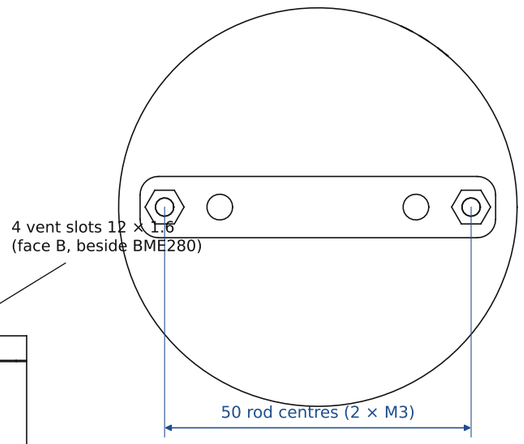


FRONT (from +Y, face A)

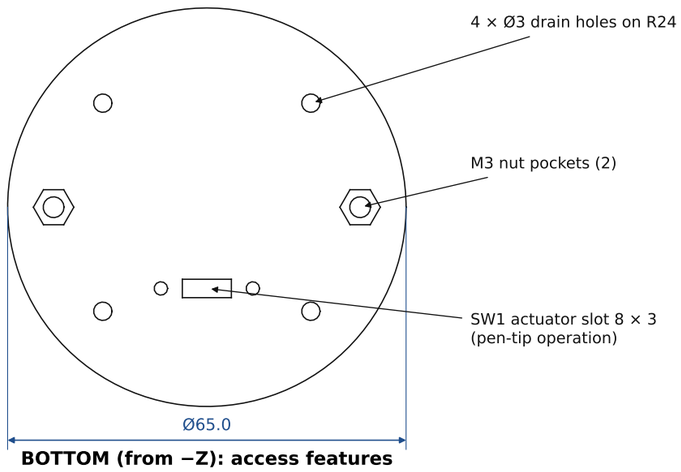
dash-dot outline = reference envelope Ø66 × 115 (assumed)



RIGHT SIDE (from +X)



TOP (from +Z)



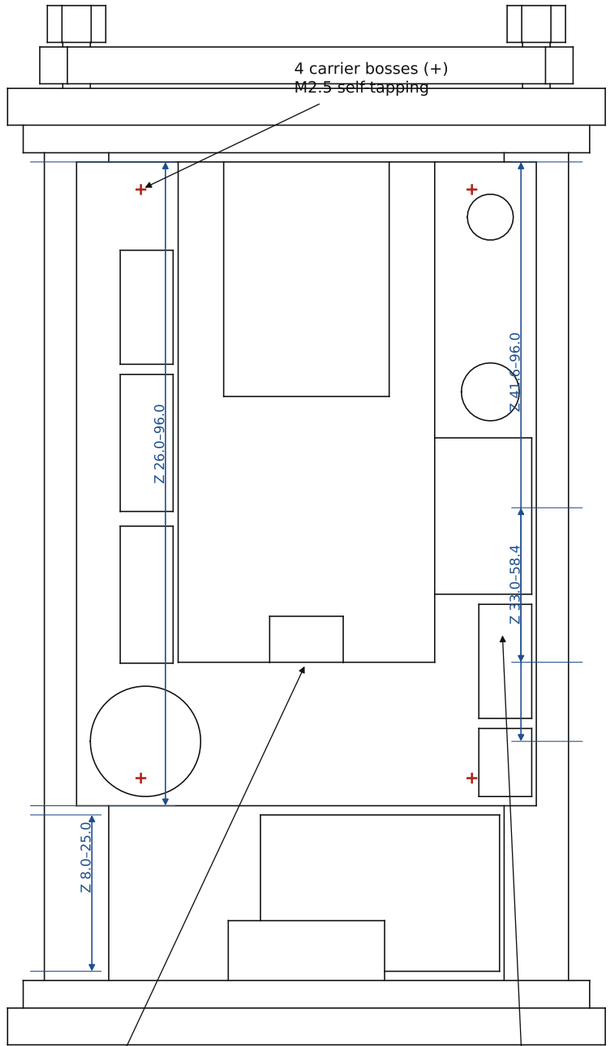
BOTTOM (from −Z): access features

STRATOS — General arrangement (exterior)

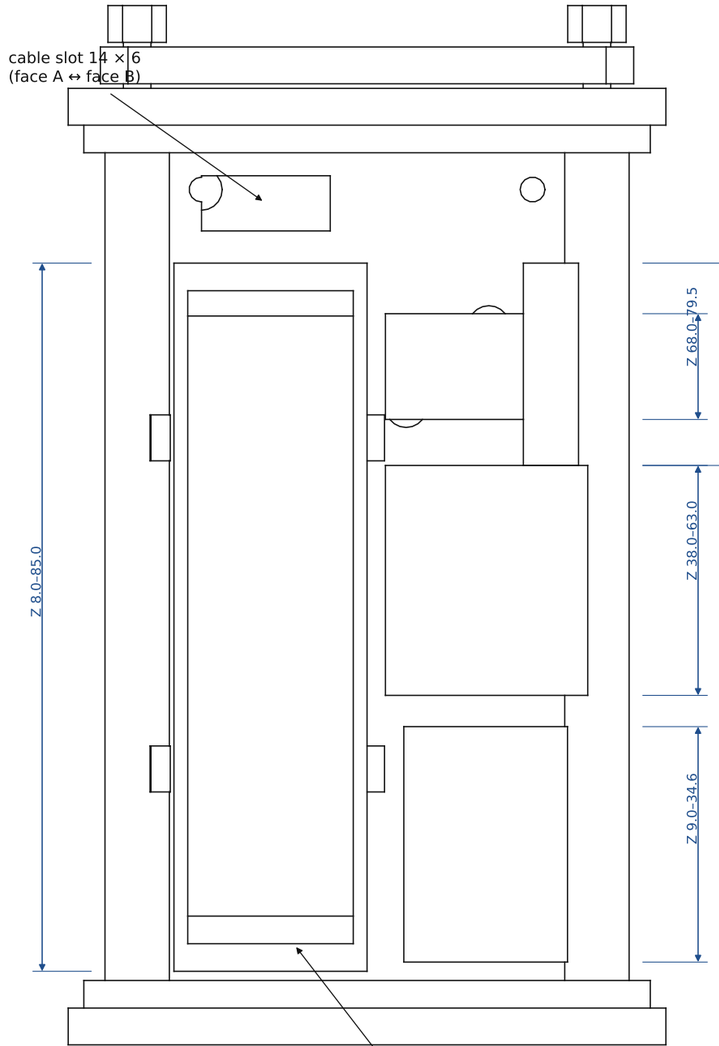
Team Nothingburger · Rev A · 2026-10-24 · scale 1:1 · units mm

Sheet 1 of 4 · generated from cad/stratos_cad.py (HLR projection)

PROPOSED DESIGN — NOT YET FABRICATED. Values marked * are PRELIMINARY (module outlines are estimates to be measured on the actual parts when received).

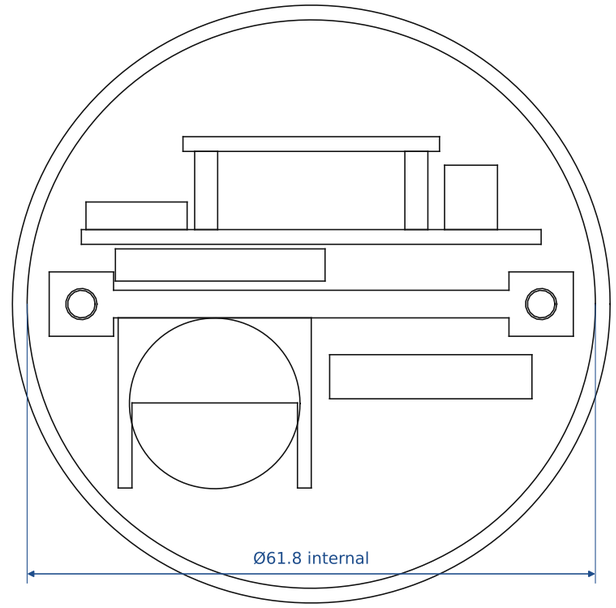


FACE A — controller side (from +Y)



FACE B — battery / sensors / radio (from −Y)

SECTION Z = 55 (scale 1.5:1, from +Z)



face A (+Y) up · face B (−Y) down

Smallest computed clearances (CAD check, cad_report.json)

Item	Radial clearance to shell bore (mm)
BUZZER	1.54
Batt_Holder	1.90
J1_BATT	2.12
J4_SWITCH	2.12
Spine	2.19
Regulator	4.00
Carrier	4.62
LoRa	4.78
J2_SENS	5.64

ESP32 micro-USB faces −Z:
reached with shell removed

U5 microSD under the carrier;
card at the −X edge (shell removed)

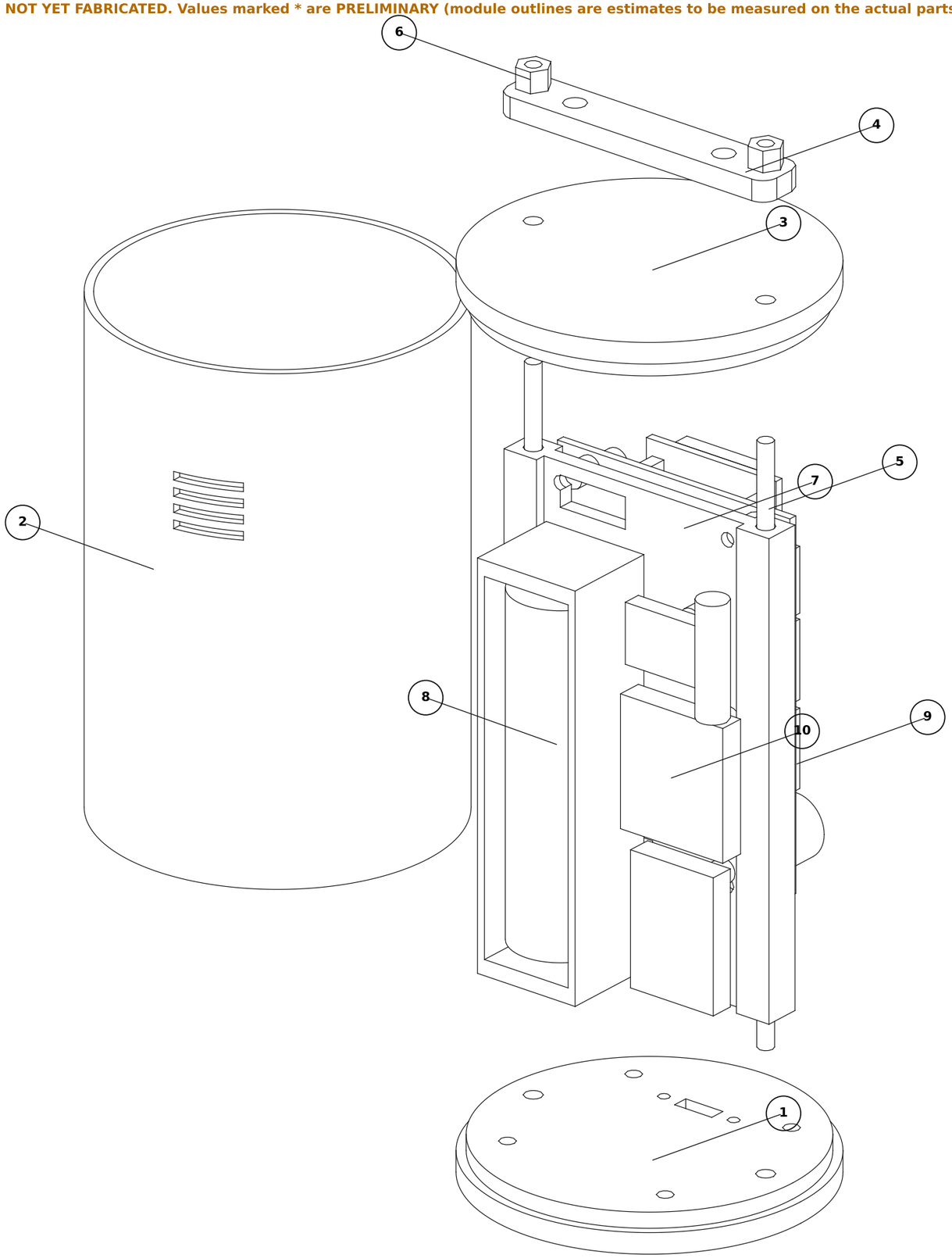
cable slot 10 × 5
(cell → U7)

STRATOS — Internal arrangement (shell removed) and section

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Sheet 2 of 4 · generated from cad/stratos_cad.py (HLR projection)

PROPOSED DESIGN — NOT YET FABRICATED. Values marked * are PRELIMINARY (module outlines are estimates to be measured on the actual parts when received).



Parts list (mechanical)

Item	Part	Material / source	Qty
1	Bottom cap	PETG, printed	1
2	Shell sleeve	PETG, printed	1
3	Top cap	PETG, printed	1
4	Recovery yoke 58×10×4	Al 6061-T6 flat bar, drilled	1
5	Threaded rod M3 × 112.5	A2 stainless	2
6	Nut M3 (bottom) / nyloc M3 (top)	A2 stainless	2 + 2
7	Spine	PETG, printed	1
8	18650 cell + holder	see BOM	1
9	Carrier assembly	see BOM / schematic	1
10	Face B modules	see BOM / schematic	1 set

Assembly order (proposed):

1 Mount modules on spine bosses; route harness through slots.

2 Insert rods through bottom-cap pockets and spine ribs.

3 Seat spine on the bottom cap; fit SW1 in its slot.

4 Slide shell over the spine (vents aligned with BME280).

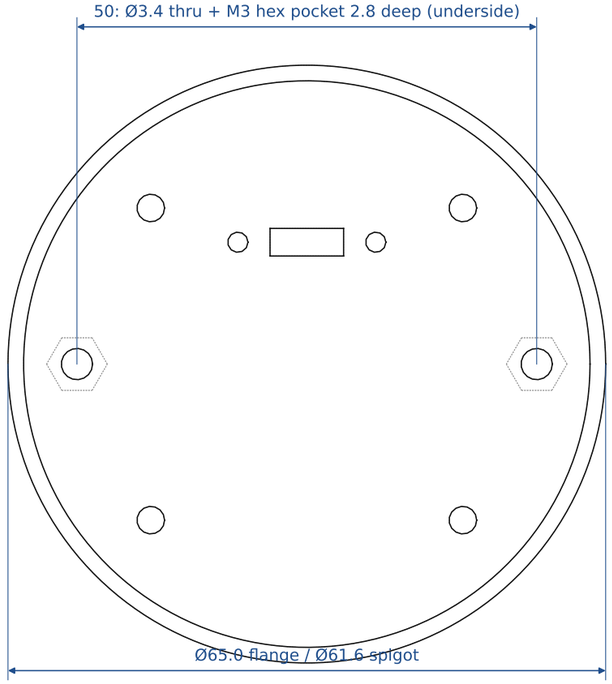
5 Fit top cap, washers, yoke and nyloc nuts; torque lightly.

6 Attach the Y-bridle (two legs) + swivel + parachute to the yoke.

STRATOS — Exploded view and parts list

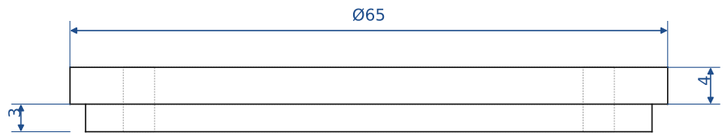
Team Nothingburger · Rev A · 2026-10-24 · isometric, not to scale · units mm

PROPOSED DESIGN — NOT YET FABRICATED. Values marked * are PRELIMINARY (module outlines are estimates to be measured on the actual parts when received).

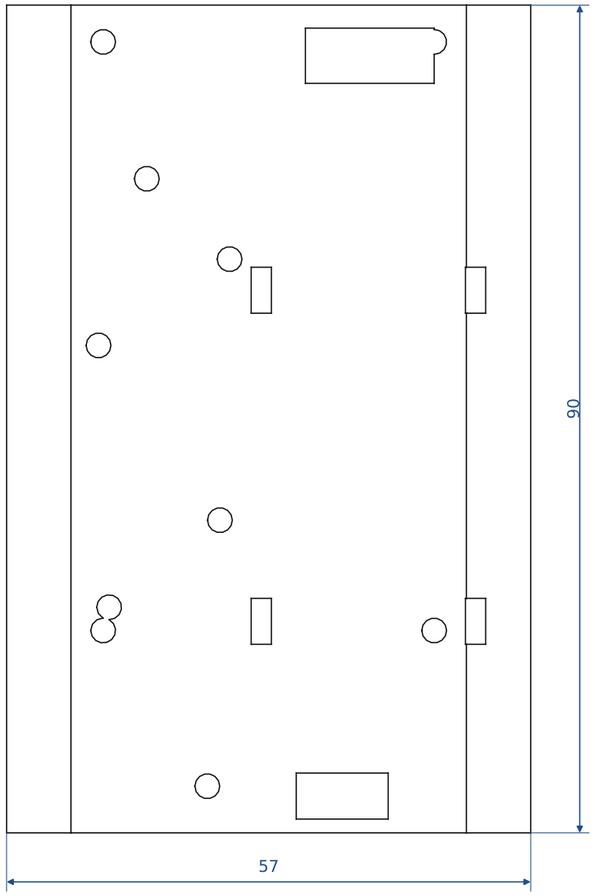


BOTTOM CAP — top view (1.5:1)

Spigot 3 mm; SW1 slot 8 × 3 with 2 × Ø2.2 at 15 mm pitch;
4 × Ø3 drains on R24. Print flange-down, 100 % perimeters at pockets.

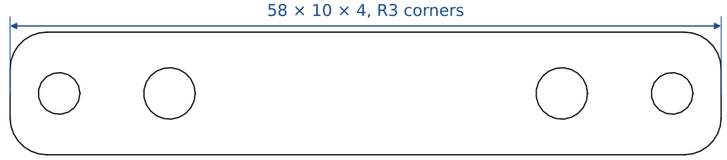


TOP CAP — front view (1.5:1)



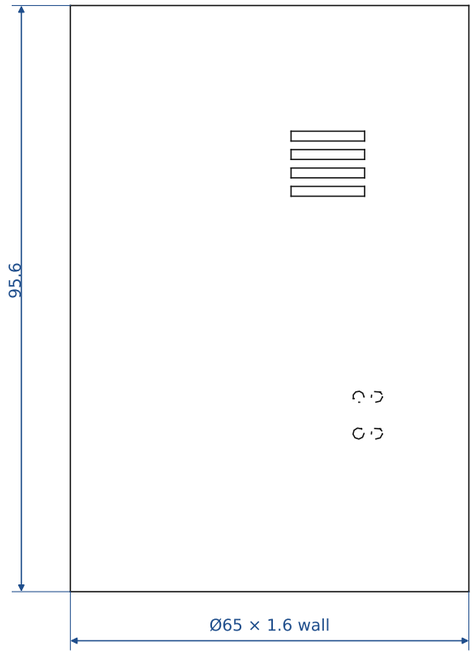
SPINE — face A view (1.5:1)

Web 3 mm; ribs 7 × 7 mm with Ø3.4 rod bores (print upright);
Ø2.7 holes / Ø6 bosses for M2.5 screws; cable slots 14 × 6 and 10 × 5;
zip-tie slots 2.2 × 5 beside the cell holder.



RECOVERY YOKE — top view (2:1)

Ø3.4 at ±25 (rods); 2 × Ø4.2 at ±16 (Y-bridle). 6061-T6 flat bar;
deburr all holes; edges broken 0.3 mm to protect the bridle.



SHELL — face B side (1:1)

Vent slots 4 × (12 × 1.6), pitch 3.0, first at Z 69.5, centred on X = 9.5;
sound holes 6 × Ø1.8 on face A at Z 30 / 36. Print upright, 3 perimeters.

STRATOS — Fabricated part details
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Sheet 4 of 4 · generated from cad/stratos_cad.py (HLR projection)